

Kundai Farai Sachikonye
Auf der Lichtung 19, 82178 Puchheim, Germany
github.com/fullscreen-triangle
kundai.sachikonye@bitspark.com
(+49) 1626386344

June 16, 2026

AMBIOM Research Group
Leibniz-Institut für Analytische Wissenschaften — ISAS — e.V.
Dortmund

Re: PhD Candidate, AMBIOM (Analysis of Microscopic BIOMedical images)
— **Reference 401_2026**

Dear Hiring Team,

I am applying for the PhD Candidate position in the AMBIOM group. The three tasks in the posting — a large-scale microscopy image-analysis pipeline, a computational framework for 3D shape modelling, and AI-based representation learning for cell modelling — describe almost exactly the work I have been doing, and I would be glad to develop it into a doctorate at ISAS.

Microscopy image analysis and computer vision. This is my core. I have built a quantitative microscopy image-analysis framework covering segmentation, point-spread-function deconvolution, registration, and morphometry, with uncertainty quantification, for fluorescence and phase-contrast microscopy. It is GPU-accelerated (WebGL/GLSL) and deployed as a live tool (hieronymus-omega.vercel.app); the underlying method is written up as a formal image-analysis paper. Extending this pipeline to large-scale 3D microscopy is a direct continuation of what I already do.

Shape modelling and representation learning. My image-analysis work centres on recovering metric-preserving geometry from image observations — scale fields, coordinate fields, geodesic distance — which is the natural basis for quantitative 3D shape characterisation. On the learning side, I work in PyTorch with CNNs, autoencoders and representation learning, and foundation-model fine-tuning, applied to imaging and spectral data and to data-driven cell- and disease-state modelling. Building a representation-learning framework for cell modelling is squarely in this line.

Biology and communication. I hold an MSc in Computational Biology and an MSc in Pharmaceutical Biotechnology, with a BSc in Biochemistry and Cell Biology, so the biological context of cell modelling is familiar ground rather than a hurdle. I write and publish regularly, and I am used to presenting technical work to interdisciplinary audiences. My degree qualifies me directly (MSc in computational biology); English is native; and I am based in Germany.

I am proactive and independent by temperament — most of the work above was self-directed — and I would value the structure, supervision, and state-of-the-art imaging infrastructure that a PhD at ISAS offers to turn it into rigorous, publishable research. I would welcome the opportunity to discuss the position.

Yours sincerely,

Kundai Farai Sachikonye